

LECTURE L5-6 • MODULE 1 • CO1

Unstructured data and ethics

Week 3 | Slides: DATA209_Module_1_Statistical_and_Data_Foundations.pptx, pages 31–37 | Laboratory: P5-6

TOPICS COVERED

- Image as a matrix; audio as a waveform
- Video as a sequence of frames
- Curse of dimensionality — introduction
- Data bias and ethical concerns

Everything becomes an array

- **Image** — a grid of pixel values. Grayscale is `(height, width)`; colour is

`(height, width, 3)`. Explore the intensity histogram: exposure, contrast, and clipping at 0 or 255 tell you what was lost at capture and cannot be recovered.

- **Audio** — amplitude sampled over time: a 1-D array plus a **sample rate**. Explore the

waveform for structure, the amplitude histogram for symmetry, and short-term energy for loudness over time. Silence is often the majority of a field recording.

- **Video** — a sequence of frames, `(frames, height, width, channels)` plus a frame rate. Each

frame is an image, so image technique applies frame by frame.

Once represented as an array, the ordinary exploratory questions apply unchanged: what is the distribution, where are the extremes, what is missing, and is anything clipped or saturated?

The curse of dimensionality — a flag planted

As dimensions grow, data density collapses exponentially. In a unit cube, the proportion of volume lying within 0.1 of an edge is 20% in one dimension, 49% in three, and 99% in twenty — almost every point is near a boundary. Distances between points converge, so "nearest neighbour" stops carrying information, and distance-based methods (k-NN, k-means) degrade first.

This is introduced here only. The response — feature selection and dimensionality reduction — is Module 4.

Bias enters before you write any code

Four kinds, all present before analysis begins:

- **Historical bias** — the world the data records was already unequal. Faithful data can encode unfair outcomes.
- **Representation bias** — some groups are under-sampled; findings will not transfer to them.

- **Measurement bias** — the proxy is not the thing. Arrests are not crime; clicks are not interest.
- **Aggregation bias** — one model for everyone hides subgroup differences. Simpson's paradox is

the statistical form of this, and it appears again in L11-12.

An ethics checklist

Run this before analysing, not after:

1. Who is in this dataset, who is missing, and would they have consented to this use?
2. Does it contain personal or sensitive attributes, and do you actually need them?
3. Could a finding here be used to disadvantage a group represented in the data?
4. Are proxy variables standing in for protected attributes?
5. What claim would be unsupportable outside the sampling frame — and have you written it down?

The Pima dataset records women of Pima heritage aged 21 and over. Every conclusion drawn from it is conditional on that frame, and a model trained on it must not be presented as a general screener.

KEY TERMS

Modality Sample rate Pixel intensity Clipping Curse of dimensionality
Historical bias Representation bias Proxy variable Aggregation bias

COMMON ERRORS TO AVOID

- Reporting image statistics without stating shape, dtype and value range
- Treating a proxy variable as if it were the quantity of interest
- Presenting a model trained on one population as generally applicable
- Leaving the ethics review to the end of the project

READINGS — ALL FREE TO ACCESS

1. scikit-image — images as NumPy arrays
https://scikit-image.org/docs/stable/user_guide/numpy_images.html
2. Wilke — Fundamentals of Data Visualization
<https://clauswilke.com/dataviz/>
3. Berkeley Data 100 — course notes
<https://ds100.org/course-notes/>

TEST YOURSELF

1. An 8-bit grayscale image has 12% of pixels at exactly 255. What happened, and can it be undone?
2. Why do distance-based methods degrade first in high dimensions?
3. Give an example of a proxy variable that could stand in for a protected attribute.
4. State one claim you could not defend from the Pima dataset.